Fundamentals of Data Science

Assignment 2

Name: Praptee Dewan

Student ID: 25068590

Module Code: UFCFK1-15-0

Instructor: Kaushal Rimal

Submission Date: 17 April 2026

Github link: https://github.com/dprapti25-coder/FODS-Submissions.git

1) **Function for Arithmetic Operations**

This program defines a function that takes two integers and calculates sum, difference, product, and quotient.

- **Code:**

```
def calculate(a, b):
    print("Sum:", a + b)
    print("Difference:", a - b)
    print("Product:", a * b)

    if b != 0:
        print("Quotient:", a / b)
    else:
        print("Quotient: Cannot divide by zero")

num1 = int(input("Enter first integer: "))
num2 = int(input("Enter second integer: "))

calculate(num1, num2)
```

- **Output Screenshot:**

```
Enter first integer: 5
Enter second integer: 20
Sum: 25
Difference: -15
Product: 100
Quotient: 0.25
```

2) **Separate Functions**

This program implements separate functions for arithmetic operations such as addition, multiplication, division, floor division, and exponentiation.

- **Code:**

```
def add(a, b):
    return a + b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b != 0:
        return a / b
    else:
        return "Cannot divide by zero"

def floor_divide(a, b):
    if b != 0:
        return a // b
    else:
        return "Cannot divide by zero"

def power(a, b):
    return a ** b

num1 = int(input("Enter first number: "))
num2 = int(input("Enter second number: "))

print("\n--- Results ---")
print("Addition:", add(num1, num2))
print("Multiplication:", multiply(num1, num2))
print("Division:", divide(num1, num2))
print("Floor Division:", floor_divide(num1, num2))
print("Exponentiation:", power(num1, num2))
```

- **Output Screenshot:**

```
Enter first number: 24
Enter second number: 67

--- Results ---
Addition: 91
Multiplication: 1608
Division: 0.3582089552238806
Floor Division: 0
Exponentiation: 297956726790367764888892739590339738467386455378146161589664867501235719127323045553
372135424
```

3) Armstrong Number

This program checks whether a number is an Armstrong number using a function.

- **Code:**

```
def is_armstrong(num):
    power = len(str(num))
    total = sum(int(d)**power for d in str(num))
    return total == num

num = int(input("Enter number: "))
print("Armstrong" if is_armstrong(num) else "Not Armstrong")
```

- **Output Screenshot:**

```
Enter number: 45
Not Armstrong
```

4) Count Occurrences (Dictionary)

This function takes a list and returns a dictionary showing frequency of each element.

- **Code:**

```
def count_frequency(numbers):
    freq = {}  # empty dictionary

    for num in numbers:
```

```
        if num in freq:
            freq[num] += 1
        else:
            freq[num] = 1

    return freq

list1 = [1, 2, 2, 3, 3, 3]
print("List 1:", list1)
print("Frequency:", count_frequency(list1))
print()

list2 = [5, 5, 5, 1, 2, 2]
print("List 2:", list2)
print("Frequency:", count_frequency(list2))
print()

list3 = [10, 20, 10, 30, 20, 10]
print("List 3:", list3)
print("Frequency:", count_frequency(list3))

list4 = [25, 20, 20, 25, 20, 3]
print("List 4:", list3)
print("Frequency:", count_frequency(list4))
```

- **Output Screenshot:**

```
List 1: [1, 2, 2, 3, 3, 3]
Frequency: {1: 1, 2: 2, 3: 3}

List 2: [5, 5, 5, 1, 2, 2]
Frequency: {5: 3, 1: 1, 2: 2}

List 3: [10, 20, 10, 30, 20, 10]
Frequency: {10: 3, 20: 2, 30: 1}
List 4: [10, 20, 10, 30, 20, 10]
Frequency: {25: 2, 20: 3, 3: 1}
```

5) **Reverse Alphabetical Sorting**

This function sorts student names in reverse alphabetical order.

- **Code:**

```
'''Sorting a list of names in reverse alphabetical order'''

def sort_reverse(names):
    return sorted(names, reverse=True)

students = []

n = int(input("Enter number of students: "))

for i in range(n):
    name = input("Enter student name: ")
    students.append(name)

sorted_names = sort_reverse(students)

print("\nNames in reverse alphabetical order:")
for name in sorted_names:
    print(name)
```

- **Output Screenshot:**

```
Enter student name: June
Enter student name: Meena
Enter student name: Sumila
Enter student name: Prakash
Enter student name: Sherpa
Enter student name: Micsen

Names in reverse alphabetical order:
Sumila
Sherpa
Prakash
Micsen
Meena
June
```

6) Search Country

This function searches for a country in a list and returns its index.

- **Code:**

```
def search_country(country_list, country):
    for i in range(len(country_list)):
        if country_list[i] == country:
            return i
    return "Not Found in List"

countries = []

n = int(input("Enter number of countries: "))

for i in range(n):
    name = input("Enter country name: ")
    countries.append(name)

search = input("Enter country to search: ")

result = search_country(countries, search)

if result == "Not Found in List":
    print(result)
else:
    print("Country found at index:", result)
```

- **Output Screenshot:**

```
Enter number of countries: 6
Enter country name: Nepal
Enter country name: Korea
Enter country name: Japan
Enter country name: Switzerland
Enter country name: Finland
Enter country name: Brunei
Enter country to search: Finland
Country found at index: 4
```

7) **Character Frequency**

This function counts occurrences of characters in a sentence (excluding spaces).

- **Code:**

```python
def key_value(sentence):
    freq = {}

    for char in sentence:
        if char != " ":
            if char in freq:
                freq[char] += 1
            else:
                freq[char] = 1

    return freq

text = input("Enter a sentence: ")

result = key_value(text)

print("\nCharacter frequency (excluding spaces):")
print(result)
```

- **Output Screenshot:**

```
Enter a sentence: I love to collect rocks

Character frequency (excluding spaces):
{'I': 1, 'l': 3, 'o': 4, 'v': 1, 'e': 2, 't': 2, 'c': 3, 'r': 1, 'k': 1, 's': 1}
```

8) Remove Duplicates and Sort

This program removes duplicates and sorts numbers in ascending order.

- **Code:**

```
def remove_duplicates_sort(numbers):
    unique_numbers = list(set(numbers))

    # Sort the list in ascending order
    unique_numbers.sort()

    return unique_numbers

numbers = []

n = int(input("Enter how many numbers: "))

for i in range(n):
    num = int(input("Enter number: "))
    numbers.append(num)

result = remove_duplicates_sort(numbers)

print("\nList after removing duplicates and sorting:")
print(result)
```

- **Output Screenshot:**

```
Enter how many numbers: 7
Enter number: 67
Enter number: 67
Enter number: 34
Enter number: 55
Enter number: 1
Enter number: 76
Enter number: 45

List after removing duplicates and sorting:
[1, 34, 45, 55, 67, 76]
```

9) Movies Dictionary

This program stores details of 3 movies using a dictionary.

- **Code:**

```
movies = {}

for i in range(1, 4):
    print(f"\nEnter details for Movie {i}:")

    title = input("Title: ")
    director = input("Director: ")
    year = int(input("Release Year: "))
    rating = float(input("Rating (out of 10): "))

    movies[title] = {
        "Director": director,
        "Year": year,
        "Rating": rating
    }

print("\n===== MOVIE DETAILS =====\n")

for title, details in movies.items():
    print("Title:", title)
    print("Director:", details["Director"])
    print("Release Year:", details["Year"])
    print("Rating:", details["Rating"])
    print("------------------------")
```

- **Output Screenshot:**

```
Title: Everything, Everywhere, All At Once
Director: Daniel Kwan & Daniel Scheinert
Release Year: 2022
Rating (out of 10): 7.7

Enter details for Movie 2:
Title: Past Live
Director: Celine Song
Release Year: 2023
Rating (out of 10): 7.8

Enter details for Movie 3:
Title: Close
Director: Lukas Dhont
Release Year: 2022
Rating (out of 10): 7.8

===== MOVIE DETAILS =====

Title: Everything, Everywhere, All At Once
Director: Daniel Kwan & Daniel Scheinert
Release Year: 2022
Rating: 7.7
------------------------
Title: Past Live
Director: Celine Song
Release Year: 2023
Rating: 7.8
------------------------
Title: Close
Director: Lukas Dhont
Release Year: 2022
Rating: 7.8
```

10) Dice Guessing Game

This program simulates a dice guessing game with conditions for correct and near guesses.

- **Code:**

```
import random

dice = random.randint(1, 6)

guess = int(input("Guess the dice value (1 to 6): "))
```

```python

print("\nDice value was:", dice)

if guess == dice:
    print("😊 Correct guess! Well done!")
elif abs(guess - dice) == 1:
    print("😵 Close guess! You were off by 1.")
else:
    print("❌ Wrong guess! Try again.")
```

- **Output Screenshot:**

```
Guess the dice value (1 to 6): 5

Dice value was: 1
❌ Wrong guess! Try again.
```

```
Guess the dice value (1 to 6): 5

Dice value was: 1
❌ Wrong guess! Try again.
```